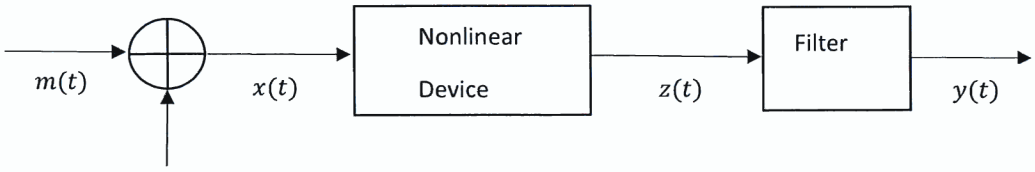
NAME: **NUMBER:**

ELEC 361 Analog Communication Systems Bonus Quiz **Fall 2023**
Prof. Dr. Oğuz Kucur **Time: 30 min.** **Jan. 5, 2024**

30 p.

Q (25 p.): The message signal $m(t)$ with bandwidth B Hz is applied to the modulator shown in the figure below ($f_C \gg B$). The nonlinear device has the input-output relationship $z(t) = 10.a.x(t) + b.x^2(t)$, where a, b are constants taking values 0 or 1. a) Find $z(t)$ and $Z(f)$. For each following case, find the vaules of a and b, specify the filter, and express $y(t)$: i) DSB-SC-AM, ii) DSB-LC-AM, iii) Upper SSB-AM. b) For case ii, if $f_C = 1\ kHz$ and $m(t) = 4Cos(200\pi t)$, find the modulation index and efficiency. Find $Y(f)$. Sketch $y(t)$ and $Y(f)$.



a) $z(t) = \underbrace{10.a.m(t)}_{LP} + \underbrace{10.a.Cos(2\pi f_c t)}_{BP\ at\ f_c} + \underbrace{b.m^2(t)}_{LP} + \underbrace{2b\,m(t)Cos(2\pi f_c t)}_{BP\ at\ f_c} + \underbrace{\frac{b}{2}}_{LP} + \underbrace{\frac{b}{2}Cos(2\pi 2f_c t)}_{BP\ at\ 2f_c}$

$Z(f) = 10a\,M(f) + 5a[\delta(f-f_c)+\delta(f+f_c)] + b[M(f)\circledast M(f)] + b[M(f-f_c)+M(f+f_c)] + \frac{b}{2}\delta(f) + \frac{b}{4}[\delta(f-2f_c)+\delta(f+2f_c)]$

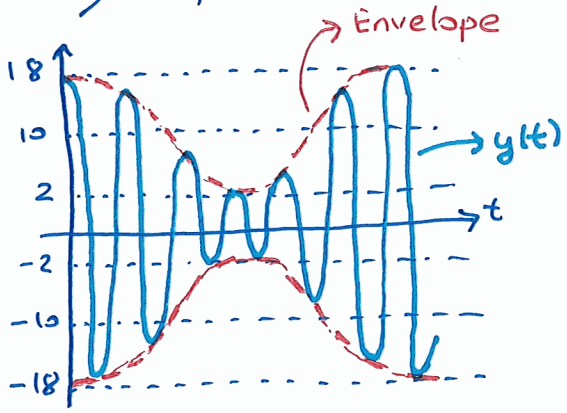
i) DSB-SC-AM: $a=0, b=1$, BPF: Gain 1, C.F. $= f_c$, BW $= 2B$ Hz $\Rightarrow y(t) = 2.m(t)Cos(2\pi f_c t)$

ii) DSB-LC-AM: $a=1, b=1$, Filter the same as i), $y(t) = 10Cos(2\pi f_c t) + 2m(t)Cos(2\pi f_c t)$
$= 10[1 + \frac{2}{10}m(t)]Cos(2\pi f_c t)$

iii) U-SSB: $a=0, b=1$, BPF: Gain 1, C.F. $= f_c + \frac{B}{2}$, BW $= B$ Hz, i.e.,
Gain 1 if $f_c \le |f| \le f_c + B \Rightarrow y(t) = 2[m(t)Cos(2\pi f_c t) - \hat{m}(t)Sin(2\pi f_c t)]$

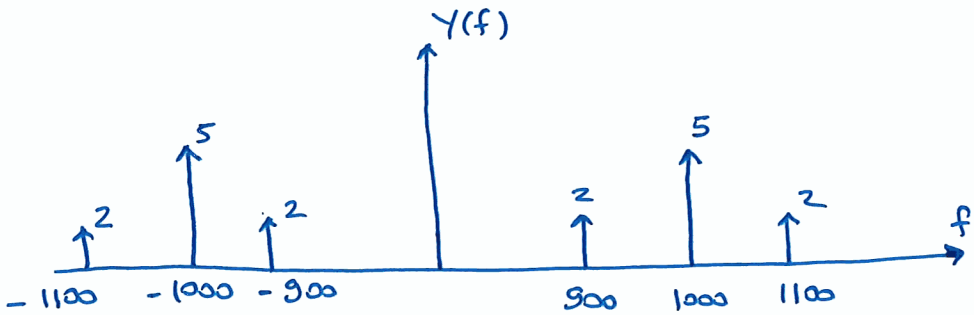
b) $y(t) = 10[1 + \underbrace{\frac{2}{10}}_{\mu}\,4\,\underbrace{Cos(200\pi t)}_{m_n(t)}]Cos(2000\pi t)$

$\Rightarrow \mu = 0,8$ Sinusoidal message signal (tone mod.) $\Rightarrow V = \frac{\mu^2}{2+\mu^2} = 0,242 \simeq 24\%$



$y(t) = 10Cos(2\pi 1000t) + 4[Cos(2\pi 900t) + Cos(2\pi 1100t)]$

$Y(f) = 5[\delta(f-1000)+\delta(f+1000)] + 2[\delta(f-900)+\delta(f+900)] + 2[\delta(f-1100)+\delta(f+1100)]$

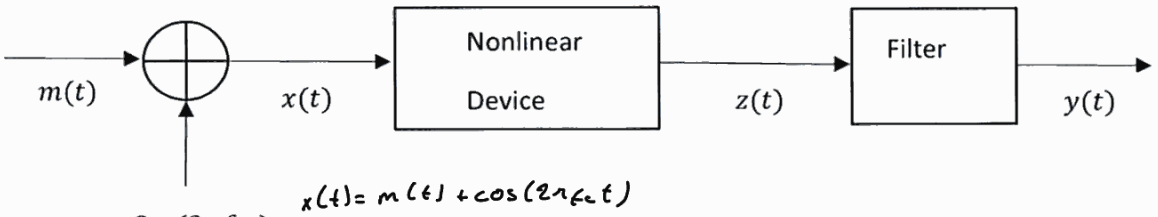


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$Cos(2\pi f_C t)$

$x(t) = m(t) + \cos(2\pi f_c t)$

a) $z(t) = 10\,a\,x(t) + b\,x^2(t)$

$\cos^2(2\pi f_c t) = \frac{1}{2} + \frac{1}{2}\cos(2\pi 2 f_c t)$

$= 10a\,m(t) + 10a\cos(2\pi f_c t) + b\,m^2(t) + 2b\,m(t)\cos(2\pi f_c t) + \frac{1b}{2} + \frac{1b}{2}\cos(2\pi 2 f_c t)$

i) $a=0 \quad b=1 \Rightarrow z(t) = m^2(t) + 2m(t)\cos(2\pi f_c t) + \frac{1}{2} + \frac{1}{2}\cos(2\pi 2 f_c t)$

BPF at f_c $BW = 2B$ $y(t) = 2m(t)\cos(2\pi f_c t)$

ii) $a=1 \quad b=1$

$z(t) = 10\,m(t) + 10\cos(2\pi f_c t) + m^2(t) + 2\,m(t)\cos(2\pi f_c t) + \frac{1}{2} + \frac{1}{2}\cos(2\pi 2 f_c t)$

BPF at f_c $BW = 2B$ $y(t) = 10\cos(2\pi f_c t) + 2m(t)\cos(2\pi f_c t)$

$y(t) = 10\left[1 + \frac{2m(t)}{10}\right]\cos(2\pi f_c t)$

iii) $a=0 \quad b=1 \Rightarrow$

$z(t) = m^2(t) + 2m(t)\cos(2\pi f_c t) + \frac{1}{2} + \frac{1}{2}\cos(2\pi 2 f_c t)$

BPF at $f_c + \frac{B}{2}$ $BW = B$

$y(t) = 2\left[m(t)\cos(2\pi f_c t) - m_h(t)\sin(2\pi f_c t)\right]$

b) $f_c = 1k \quad m(t) = 4\cos(2\pi 100 t) \quad m_p = 4 \quad m_n(t) = \cos(2\pi 100 t)$

$y(t) = 10\left[1 + \frac{8}{10}\underbrace{m_n(t)}_{\cos(2\pi 100 t)}\right]\cos(2\pi 1k t) \quad \mu = 0{,}8$

$= 10\cos(2\pi 1k t) + 8\left[\cos(2\pi 100 t)\cdot\cos(2\pi 1k t)\right]$

$= 10\cos(2\pi 1k t) + 4\cos(2\pi 1{,}1k t) + 4\cos(2\pi 0{,}9k t)$

$Y(f) = 5\left[\delta(f+1k) + \delta(f-1k)\right] + 2\left[\delta(f+1{,}1k) + \delta(f-1{,}1k) + \delta(f+0{,}9k) + \delta(f-0{,}9k)\right]$

$V = \frac{\mu^2 P_{mn}}{1+\mu^2 P_{mn}}$ $P_{mn} = \frac{1}{2}$ Ton modülasyonu

$= \frac{\mu^2}{2+\mu^2}$

$= \%24$

